



University of Frontier Technology, Bangladesh

Department of Internet of Things and Robotics Engineering

Assignment-1 Solution

Course Code: IRE 315

Course Title: Embedded Systems

Submitted To-

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Answer to the question No. 1

Given,

Task i	P_i	D_i	C_i
P_1	6	6	1
P_2	8	8	2
P_3	12	12	2.5
P_4	24	24	3

1 (i)

Hyperperiod, $H = \text{LCM}(6, 8, 12, 24) = 24$

Possible divisors of 24: 1, 2, 3, 4, 6, 8, 12, 24

constraint 1: frame size must not exceed the smallest period,
 $f \leq \min(P_i)$

$$f \leq 6$$

So, possible, 1, 2, 3, 4, 5, 6

constraint 2: Each task must fit inside one frame.

$$f \geq \max(C_i)$$

$$f \geq 3$$

So, candidate frame sizes, $f = 3, 4, 6$

constraint 3: Deadline constraint
 $2f - \gcd(p_i, f) \leq d_i$

for $f=3$,

$$P_1: 2(3) - \gcd(6, 3) = 6 - 3 = 3 \leq 6$$

$$P_2: 2(3) - \gcd(8, 3) = 6 - 1 = 5 \leq 8$$

$$P_3: 6 - \gcd(12, 3) = 6 - 3 = 3 \leq 12$$

$$P_4: 6 - \gcd(24, 3) = 6 - 3 = 3 \leq 24$$

So, $f=3$ is valid.

for $f=4$,

$$P_1: 2(4) - \gcd(6, 4) = 8 - 2 = 6 \leq 6$$

$$P_2: 2(4) - \gcd(8, 4) = 8 - 4 = 4 \leq 8$$

$$P_3: 2(4) - \gcd(12, 4) = 8 - 4 = 4 \leq 12$$

$$P_4: 8 - \gcd(24, 4) = 8 - 4 = 4 \leq 24$$

So, $f=4$ is valid.

for $f=6$,

$$P_2: 2(6) - \gcd(8, 6) = 12 - 2 = 10$$

$$10 \leq 8$$

False,

So, $f=6$ is not valid.

Therefore, all valid frame sizes are,
 $f=3, 4$

1(ii)

from part (i) valid frame sizes are,

$$f=3 \text{ \& } f=4$$

I choose

$$f=4$$

Justification

frame size $f=4$ is suitable because,

1. It satisfies constraint 1:

$$4 \leq 6$$

2. It satisfies constraint 2:

$$4 \geq 3$$

3. It satisfies deadline constraints for all tasks,

$$2f - \gcd(f, p_i) \leq d_i$$

4. It gives fewer frames than $f=3$, so scheduling overhead and timer interrupt overhead are reduced.

Hyperperiod: $\text{LCM}(6, 8, 12, 24) = 24$

for $f=4$, $\frac{24}{4} = 6$

So, only 6 frames are needed in one hyperperiod.

On the other hand, for $f=3$,

$$\frac{24}{3} = 8 \text{ frames}$$

So, for $f=3$, 8 frames are needed in one hyperperiod.

Since, $f=4$ gives fewer frames, it reduces timer interrupts and scheduling overhead. Therefore, $f=4$ is the more suitable frame size.

So, suitable frame size is $f=4$.

Answer to the question No. 2

Given,

Task	Computation Time (C)	Period (T)
τ_1	1	4
τ_2	2	10

2(i)

Computing U_p ,

$$\begin{aligned} U_p &= \sum \frac{C_i}{T_i} \\ &= \frac{1}{4} + \frac{2}{10} \\ &= 0.45 \end{aligned}$$

2(ii)

Maximum server bandwidth U_s .

$$U_p + U_s \leq 1$$

$$0.45 + U_s \leq 1$$

$$U_s \leq 0.55$$

So, maximum server bandwidth $U_s^{\max} = 0.55$

2 (iii)

Computing TBS deadline,

Given, $U_s = 0.30$

Aperiodic Jobs :

$$\tau_1 : r_1 = 1, c_1 = 0.6$$

$$\tau_2 : r_2 = 3, c_2 = 0.9$$

For τ_1 ,

$$d_i = \max(r_i, d_{i-1}) + \frac{c_i}{U_s}$$

$$d_1 = \max(1, 0) + \frac{0.6}{0.3}$$
$$= 3$$

For τ_2 ,

$$d_2 = \max(r_2, d_1) + \frac{c_2}{U_s}$$

$$= \max(3, 3) + \frac{0.9}{0.3}$$

$$= 6$$

Scheduling intuition under EDF:

Now, we have:

→ Periodic jobs of τ_1 & τ_2 released at their periods

- Aperiodic τ_1 ($c=0.6, d=3$)

- Aperiodic τ_2 ($c=0.9, d=6$)

EDF will at each time choose the job with the earliest absolute deadline among all readying jobs.

Answer to the question No. 3

Given,

Task	C_i	T_i	D_i
τ_1	3	10	10
τ_2	4	15	15
τ_3	8	30	30

RM priority rule: shorter period $\rightarrow$ higher priority

So, priority order: $\tau_1 > \tau_2 > \tau_3$

3(i)

Total utilization, $U = \sum \frac{C_i}{T_i}$

$$= \frac{3}{10} + \frac{4}{15} + \frac{8}{30}$$
$$= 0.8334$$

3(ii)

Liu & Layland Bound for $n=3$

$$U_{LL} = n(2^{1/n} - 1)$$
$$= 3(2^{1/3} - 1)$$
$$= 0.779$$

Comparing $U \leq U_{LL}$

$$0.8334 > 0.779$$

So, utilization fails.

Rms utilization does not guarantee schedulability.
 But, this does not mean unschedulable. So, we use
 RTA.

3 (iii)

Response Time Analysis,

$$R_i^{k+1} = C_i + \sum_{j \in hp(i)} \left\lceil \frac{R_j^k}{T_j} \right\rceil C_j$$

For τ_1 ,

Highest priority, so no interference,

$$R_1 = C_1 = 3$$

checking schedulability, $R_1 = 3 \leq D_1 = 10$

So, τ_1 schedulable.

For τ_2 ,

Higher priority task τ_1

$$R_2^0 = C_2 = 4$$

$$R_2^1 = 4 + \left\lceil \frac{4}{10} \right\rceil 3 = 7$$

$$R_2^2 = 4 + \left\lceil \frac{7}{10} \right\rceil 3 = 7$$

So, $R_2 = 7$

$$R_2 = 7 \leq D_2 = 15$$

τ_2 schedulable.

For τ_3 ,

Higher priority tasks τ_1, τ_2

hence,

$$R_3^0 = C_3 = 8$$

$$R_3^1 = 8 + \left\lceil \frac{8}{10} \right\rceil 3 + \left\lceil \frac{8}{15} \right\rceil 4$$

$$= 8 + 1 \times 3 + 1 \times 4$$

$$= 15$$

$$R_3^2 = 8 + \left\lceil \frac{15}{10} \right\rceil 3 + \left\lceil \frac{15}{15} \right\rceil 4$$

$$= 8 + 2 \times 3 + 1 \times 4$$

$$= 18$$

$$R_3^3 = 8 + \left\lceil \frac{18}{10} \right\rceil 3 + \left\lceil \frac{18}{15} \right\rceil 4$$

$$= 8 + 2(3) + 2(4)$$

$$= 22$$

$$R_3^4 = 8 + \left\lceil \frac{22}{10} \right\rceil 3 + \left\lceil \frac{22}{15} \right\rceil 4$$

$$= 8 + 3 \times 3 + 2 \times 4$$

$$= 25$$

$$R_3^5 = 8 + \left\lceil \frac{25}{10} \right\rceil 3 + \left\lceil \frac{25}{15} \right\rceil 4$$

$$= 8 + 3 \times 3 + 2 \times 4$$

$$= 25$$

So, $R_3 = 25$

$$R_3 = 25 \leq D_3 = 25 \leq 30$$

τ_3 schedulable

So, the final answer,

Although utilization test fails,

$$0.8334 > 0.779$$

RTA gives,

$$R_1 = 3, R_2 = 7, R_3 = 25$$

And, $R_1 \leq D_1, R_2 \leq D_2, R_3 \leq D_3$

Therefore, the task set is schedulable under RM scheduling.